the oxidation state of the central metal and the charge the on the ligands attached to it. Using this relation the oxidation number can be calculated as follows

Net charge = (oxidation state of the central metal) + [(No. of ligands) X (charge on the ligand)]

Example 1:

In $[Fe(CN)_6]^{4-}$, let the oxidation number of iron is x :

The net charge: $-4 = x + 6(-1) \Rightarrow x = +2$

Example 2:

In $[Co(NH_3)_5Cl]^{2+}$, let the oxidation number of cobalt is x :

The net charge: $+2 = x + 5(0) + 1(-1) \Rightarrow x = +3$

Evaluate yourself 2:

2. In the complex, $[Pt(NO_2)(H_2O)(NH_3)_2]Br$, identify the following
   i. Central metal atom/ion
   ii. Ligand(s) and their types
   iii. Coordination entity
   iv. Oxidation number of the central metal ion
   v. Coordination number

Types of complexes:

The coordination compounds can be classified into the following types based on (i) the net charge of the complex ion, (ii) kinds of ligands present in the coordination entity.

Classification based on the net charge on the complex:

A coordination compound in which the complex ion

i. carries a net positive charge is called a cationic complex. Examples: $[Ag(NH_3)_2]^+$, $[Co(NH_3)_6]^{3+}$, $[Fe(H_2O)_6]^{2+}$, etc

ii. carries a net negative charge is called an anionic complex. Examples: $[Ag(CN)_2]^-$, $[Co(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, etc

iii. bears no net charge, is called a neutral complex. Examples: $[Ni(CO)_4]$, $[Fe(CO)_5]$, $[Co(NH_3)_3(Cl)_3]$,

Classification based on kind of ligands:

A coordination compound in which

i. the central metal ion/atom is coordinated to only one kind of ligands is called a homoleptic complex. Examples: $[Co(NH_3)_6]^{3+}$, $[Fe(H_2O)_6]^{2+}$,

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